



THE UNIVERSITY OF
MELBOURNE

COMP90050 Advanced Database Systems

Semester 1, 2026

Lecturer: Farhana Choudhury (PhD)

Live lecture – week 3





Subject information

Quiz 1 is done!

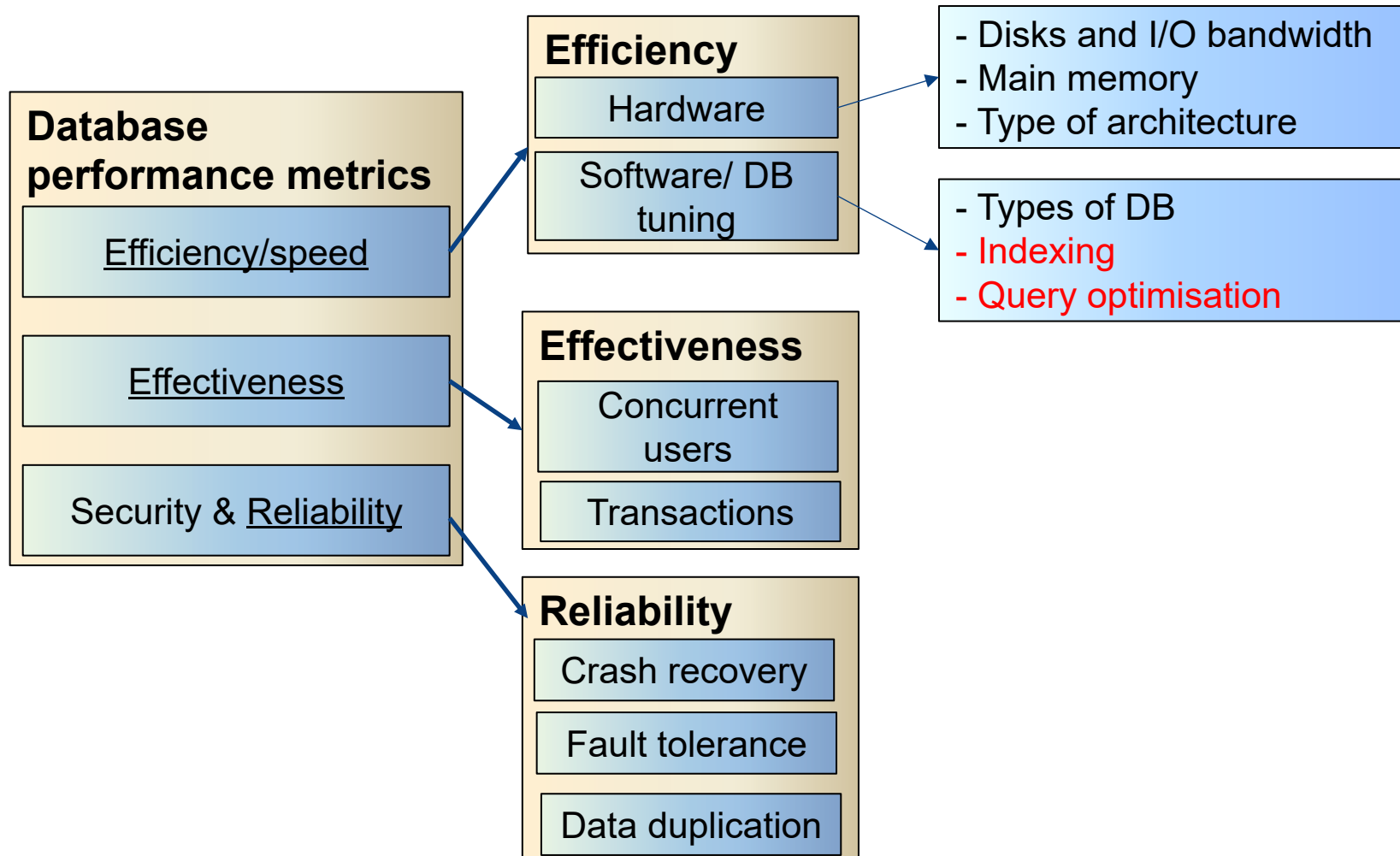
- Quizzes are helpful to keep you engaged and up to date with the subject

Practicals – during tutorials (1st one on Week 4)

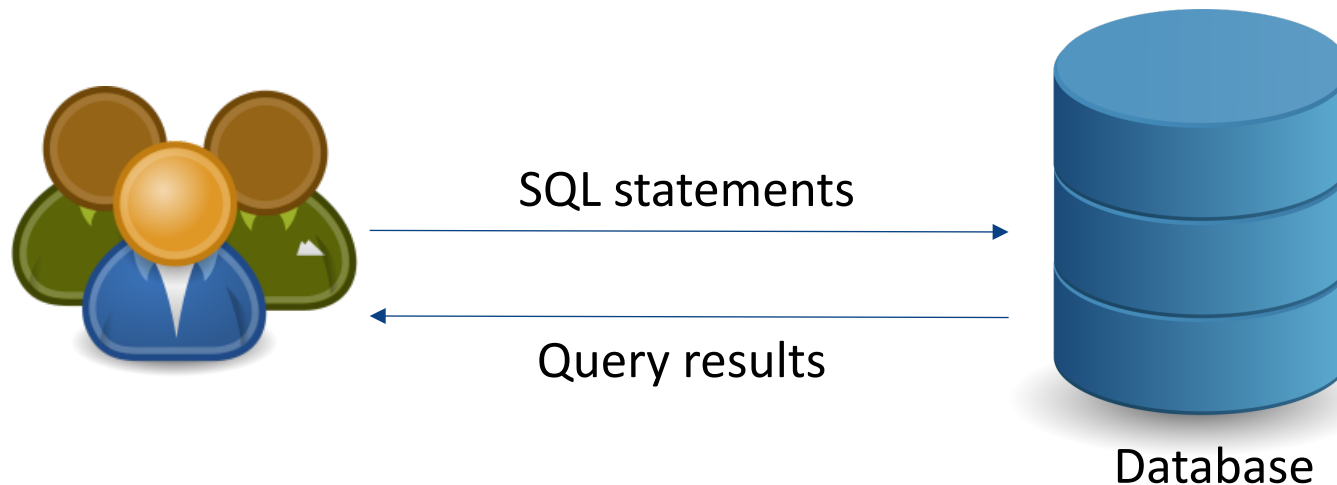


Discussion on the topics of the pre-recorded lecture and additional contents

Core Concepts of Database management system



DB queries



Efficiency is one of the most important requirements for client-facing databases

The best execution strategy to find the query results – an integral part of DBMS



Indexing is Critical for Efficiency

- An **index file** consists of records (called **index entries**) of the form **search-key, pointer to where data is**
- Index files are typically much smaller than the original data files and many parts of it are already in memory



Effect of indexes – do they always improve performance?



Time for a poll - [Pollev.com/farhanachoud585](https://pollev.com/farhanachoud585)

10101	Srinivasan	Comp. Sci.	65000	
12121	Wu	Finance	90000	
15151	Mozart	Music	40000	
22222	Einstein	Physics	95000	
32343	El Said	History	60000	

Query: Find the names of the instructors teaching 'Comp. Sci.'

Scenario 1: No indexes

Scenario 2: B+tree index on IDs

Both of the above will have the same performance



Effect of indexes – do they always improve performance?

Time for a poll - [Pollev.com/farhanachoud585](https://pollev.com/farhanachoud585)

ID	Name	Department
1	Jane	Comp. Sci
2	John	Biology
3	Alice	Comp. Sci

ID	Position	Salary
1	Lecturer	75,000
2	Research assistant	40,000
3	Senior lecturer	82,000

Query: Find the names of the instructors with salary > 80000

>0

Scenario 1: No indexes

Scenario 2: B+tree index on salary

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Scenario 1: No indexes

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Both of the above will have the same performance

Depends on statistics (e.g., rows that need to be retrieved/joined)



Search using Indexes

We have seen some example index structures

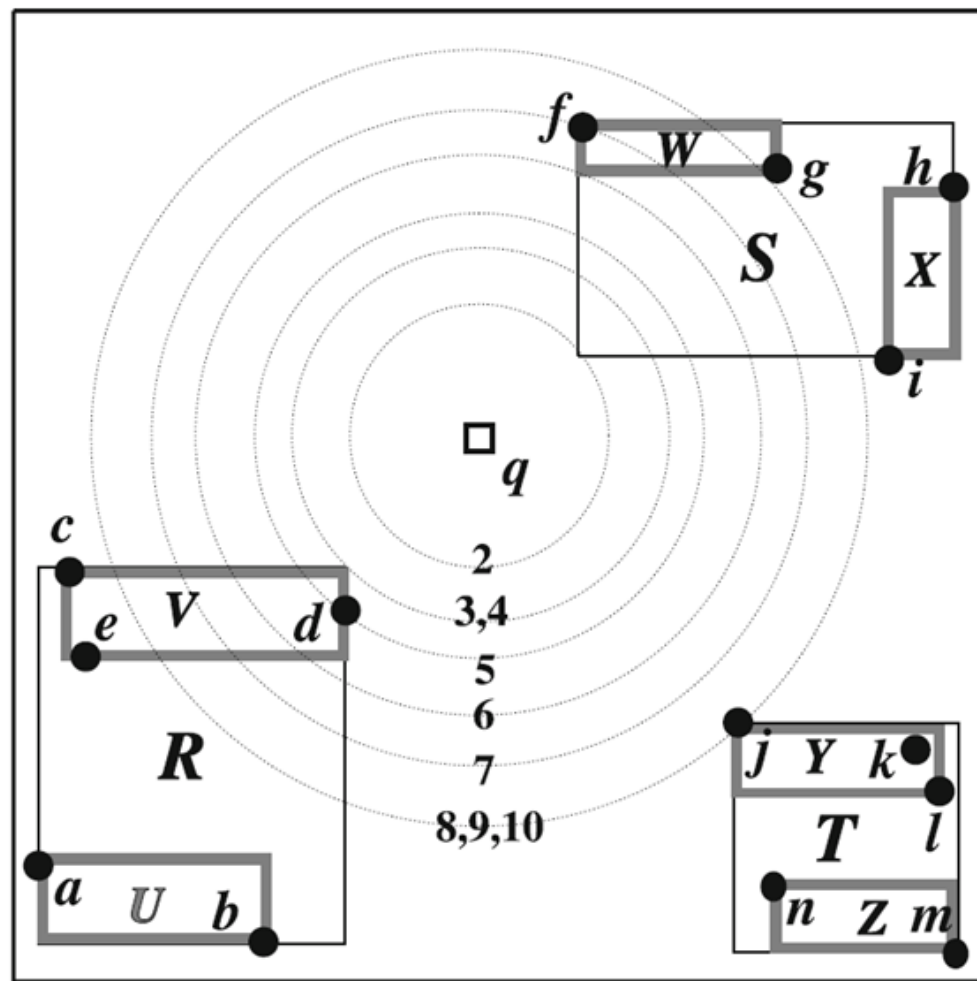
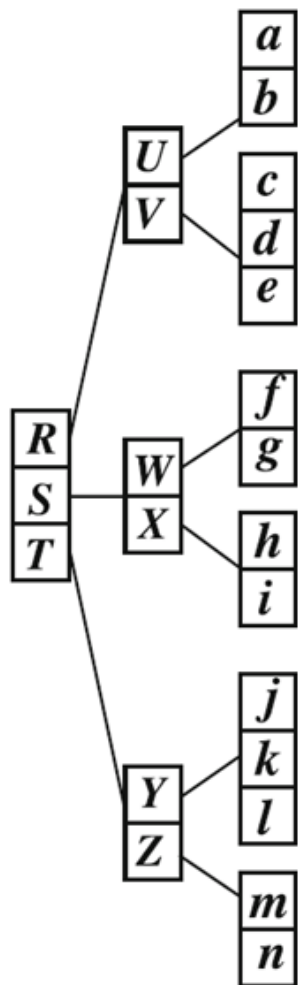
- B+ tree – finding a particular value, finding a range of values
- Hash index – finding a particular value
- Bitmap index – finding total number
- Spatial indexes – range query, nearest neighbor query
 - Quadtree
 - R-tree

Nearest neighbor in R-Trees

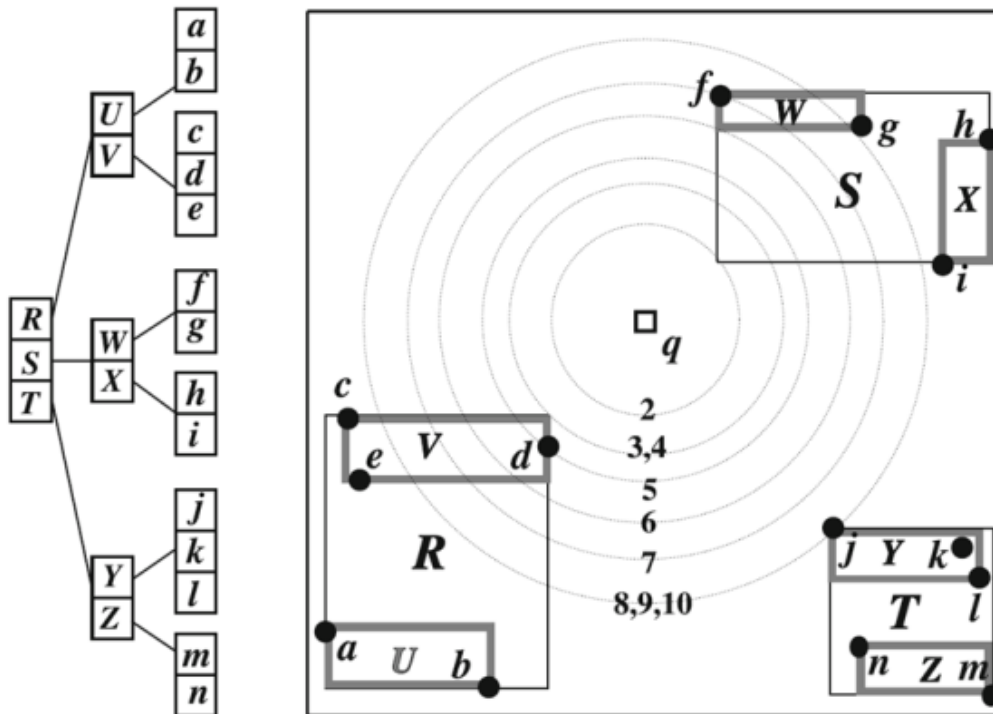
To find the nearest neighbor of a given query point/region, do the following, starting from the root node:

- Use a sorted priority queue of the R-tree nodes based on the minimum distance from the query
- Traverse the node that is in the top of the priority queue, and put its elements in the queue. Continue
- Stop when the top node is a data object (first NN has been found)
- To find k number of NN, continue the above steps until k number data objects are found as the top node of the queue.

This algorithm is a best-first search algorithm



Nearest Neighbor Query on R-tree



Why do we look at the search / query algorithms?

Step	Priority queue	Retrieved NN(s)
1	$\langle S, R, T \rangle$	$\langle \rangle$
2	$\langle R, W, T, X \rangle$	$\langle \rangle$
3	$\langle V, W, T, X, U \rangle$	$\langle \rangle$
4	$\langle d, W, T, X, c, e, U \rangle$	$\langle \rangle$

Step 5 finds d as the first NN (using Best First search)



Effect of indexes – do they always improve performance?

Group activity - <https://tinyurl.com/4x87h692>

ID	Name	Department
1	Jane	Comp. Sci
2	John	Biology
3	Alice	Comp. Sci
4	Xiaolu	Biology

ID	Position	Salary
1	Lecturer	75,000
2	Research assistant	40,000
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There are indexes constructed on:

- B+tree on IDs of both tables
- Hash index on Department
- Bitmap index on position

What will be the performance of data insertion, deletion, and updates, compared to no indexing (or just on IDs)



What you need to do now?

- Given a data set, when uploading to the DBMS
 - Find the potential query types
 - Research what indices that particular DBMS would have for that data type
 - Research for what queries you would better do on what index
 - Create index if you have large data
 - Monitor performance
 - Tune or create other indices

Summary

